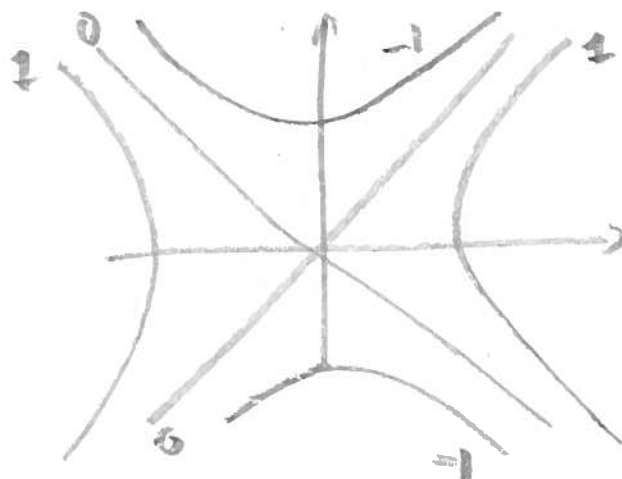


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MATHEMATICS 2210
Midterm 2, Fall semester 2012

Name: KEY

1. (15 pts) Draw the level curves for $z = x^2 - y^2$ and $k = -1, 0, 1$.



$$\begin{aligned} 1 &= x^2 - y^2 \\ 0 &= x^2 - y^2 \\ -1 &= x^2 - y^2 \end{aligned}$$

2. (15 pts) Show that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2} = 0.$$

$$\begin{array}{ccccc} 0 & \leq & \frac{x^2 y^2}{x^2 + y^2} & \leq & \frac{(x^2 + y^2) y^2}{(x^2 + y^2)} = y^2 \\ \downarrow & & & & \downarrow \\ 0 & & & & 0 \end{array}$$

So the limit is 0 by the Squeeze Theorem

3. (15 pts) Show that the following limit does not exist

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + 2y^2)}{x^2 + y^2}$$

$$x=0 \quad \lim_{y \rightarrow 0} \frac{\sin 2y^2}{y^2} = 2$$

$$y=0 \quad \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = 1$$

$2 \neq 1$ so limit DNE

4. (15 pts) Find the tangent plane to $x^2 - 2y^2 + 3z^2 = 0$ at $(1, -1, -1)$.

$$\nabla f = (2x, -4y, 6z) \quad \nabla f(1, -1, -1) = (2, 4, -6)$$

$$(2, 4, -6)(x-1, y+1, z+1) = 0$$

$$2x + 4y - 6z = 4$$

7. (10 pts) Let $x^4 - xy + y^3 = 0$. Find dy/dx by implicit differentiation.

$$\frac{dy}{dx} = - \frac{\partial F / \partial x}{\partial F / \partial y} = - \frac{4x^3 - y}{-x + 3y^2}$$

5. (15 pts) Let $f(x, y) = x^3 + xy + y^3$.

i) Compute the gradient ∇f and the directional derivative $D_{\bar{u}}f$ at $P = (2, 1)$ where $\bar{u}$ is in the direction of $(-2, 3)$

ii) Find the direction (unit vector) in which $f(x, y)$ decreases the fastest at $P = (2, 1)$. What is the rate of change in that direction?

$$a) \nabla f = (3x^2 + y, x + 3y^2) \quad \nabla f(2, 1) = (13, 5)$$

$$\bar{u} = \frac{1}{\sqrt{13}}(-2, 3) \quad D_{\bar{u}}f = (13, 5) \cdot (-2, 3) \cdot \frac{1}{\sqrt{13}} = \frac{-11}{\sqrt{13}}$$

$$ii) -\frac{1}{\sqrt{194}} \cdot (13, 5)$$

the rate of change is $-\sqrt{194}$

6. (15 pts) Let $W(x, y) = \sin(xy)$, $x = s^2 + t^2$ and $y = st$. Find $\partial W / \partial t$ using the chain rule. Express your answer in terms of s, t .

$$\frac{\partial W}{\partial t} = \frac{\partial W}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial W}{\partial y} \frac{\partial y}{\partial t} = y \cos(xy) \cdot 2t + x \cos(xy) \cdot s$$

$$= st \cos((s^2 + t^2)st) \cdot 2t + (s^2 + t^2) \cos((s^2 + t^2)st) \cdot s$$